MGM Macau and Cotai

Resource Efficiency Management Report

October 2021

Energenz Consulting Limited

EXECUTIVE SUMMARY

Energenz has been working closely with MGM to implement the EMOs as well as identify additional opportunities to achieve ISO certification energy targets. Energenz appreciates the great support and trust from MGM team over the years, in particular the Facility Management and the Sustainability Teams.

Key Performance Indicators

Energenz reports both normalized and actual savings on a monthly basis for both properties. The normalized savings are determined based on a 2019 baseline, with aligned annual savings targets to meet both ISO50001 as well as Energenz's retainer contract requirements. Please note that the reported normalized savings achieved are to-date and cumulative for each month of the respective year.

Table 1. ISO and Retainer Contract Energy Performance Targets for 2025

MGM Properties	Baseline	Energy Savings	Normalized Savings Achieved Up to October 2021	2022	2023	2024	2025
Macau	2019	Target	-4%	-8%	-12%	N/A	**-13%**
		Actual	**-6.5%**	-	-	-	-
Cotai	2019	Target	-4%	-8%	-12%	N/A	**-13%**
		Actual	**-6.2%**	-	-	-	-

**MGM Cotai ISO energy performance: e-bus consumption is included.*

The actual monthly savings are determined against the 2019 baseline consumption and cumulative for each month of 2021.

Table 2. Internal Actual Energy Performance Targets for 2025

MGM Properties	Baseline	Actual Saving Target[1]	Savings Achieved Up to October 2021
Macau	2019	15%	**-6.6%**
Cotai	2019	15%	**-4.5%**

Highlighted Changes in Operation

MGM Cotai has fully utilized Heat Pump for maintaining DHW temperatures and disinfecting the hot water system from 6 August 2021.

Steam to water heat exchanger in MGM Macau has been switched on for dehumidification at low zone guest room (L5 – L17) starting from 14 July 2021. This practice is continued in October 2021.

Hot water supply for FOH and BOH toilets has been enabled on 22 October 2021.

[1] Actual Energy Savings Target for 2025 has been established based on future CAPEX projects, following review with MGM teams

Consumption Intensity

The YTD electricity and water use intensities are tracked on a monthly basis for both properties:

Table 3. Year to Date Actual Electricity Use Intensity (EUI)

MGM Properties	YTD EUI (kWh/m^2)	% Change Compared to 2019
Macau	26.3	**-7.9%**
Cotai	27.2	**-2.9%**

Table 4. Year to Date Actual Water Use Intensity (WUI)

MGM Properties	YTD WUI (m^3/ m^2)	% Change Compared to 2019
Macau	0.21	**-14.8%**
Cotai	0.17	**-16.6%**

Note: Cotai Area: 374,138 m^2 Macau Area: 211,545 m^2

ISO 50001 Certification Audit

Macau

The ISO 50001 certificate has been received by MGM on 1 April 2021. The next internal audit will be tentatively scheduled on mid January 2022.

Cotai

The external surveillance audit has been completed on 22 October 2021 and without any "Nonconformity" or "Areas for Improvement". The next internal audit will be held on 26 November 2021.

Target Milestones for Year 2021

The following activities are planned for completion this year:

- To perform ISO 50001 2021 Internal and External Audit for MGM Macau and MGM Cotai
- To deploy and fine tune DGLux dashboard with heatmap development extension
- To complete analysis on MGM Cotai spectacle thermal comfort investigation
- To track and monitor changes in air change rates per hour for regular reporting
- To investigate and implement the below list of projects during Year 2021

Table 5. MGM Macau Year 2021 Projects

Item	Initiatives
Electricity	
1	Smoking Lounge Exhaust Fans Operation Optimization
2	Optimize Winter Operation of Cooling Tower Control Strategy - Fan Control
3	Air Change Rate (ACH) Optimization
4	Implement Motor Efficiency Upgrades (24 units)
5	Implement Static Pressure Set-Point Reset in AHU (10 units)
6	Chiller Plant Optimization: Chilled Water Differential Pressure Set-point Optimization
7	Ball Room VAV Recommissioning/Rectification Work
8	Las Vegas Room (LVR) Gaming Area VAV Recommissioning/Rectification Work
9.1	PICCV Implementation – Ball room/Foyer AHU (3 Units)
9.2	PICCV Implementation – RF AHU (2 Units)
9.3	PICCV Implementation (17 Units)
10	Neon Lighting LED Replacement
Water	
11	Optimize Existing Water Softener (Podium)

Table 6. MGM Cotai Year 2021 Projects

Item	Initiatives
Electricity	
1	Air Change Rate Optimization
2	B1 Carpark and Loading Bay Lighting Operation Schedule Optimization
3	Kitchen Exhaust Fan (KEF) Demand Control Ventilation (DCV) and Hood Control for Remaining Kitchen's (e.g. Banquet, Chun, Mansion, Coast.)
4	Optimize BOH FCU and VAV Temperature Set-Point
5	Optimize Car Park Ventilation Fans Control
6	Optimize Hotel Tower Circulation Pumps Operation
7	Chiller Plant Optimization: 1. Review Chillers Staging 2. Cooling Tower Fan Speed Optimization 3. Condensing Water Pump Speed Optimization
8	Chilled Water Differential Pressure Set-Point Reset Logic Implementation
9	B2 Domestic Hot Water Heat Exchanger Upgrade for fully utilizing the heat pump
10	Implement Static Pressure & Supply Air Temperature Set-Point Reset in AHU (48 Units)
11	PICCV Implementation (7 Units)
12	Solar PV Panel Implementation
Water	
13	Extension of CDP from ST-CD-02 to ST-CD-01 for Cooling Tower Makeup

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ACKNOWLEDGEMENT

Energenz has been working with MGM Team as the energy manager to achieve energy savings in MGM properties and feed operational intelligence into the design.

The major goal of this project is not only to realize operation energy savings but also extending the life of equipment, hence reduce CAPEX. MGM has demonstrated continuous commitment to be an environmentally and socially responsible business in the past years. In addition to that Energenz has assisted MGM Macau and Cotai to achieve several other energy management and sustainability awards and certifications, such as ISO 50001 and CEM Energy Savings Award.

Energenz appreciates the great support and trust from MGM Team over the years, especially the Facilities Management and the Sustainability Teams.

1. Portfolio Performance Summary

1.1 Electricity Use Intensity (EUI)

Macau's electricity uses intensity (EUI) in October 2021 had decreased by **4.8%** compared to same month in 2019 and decreased by **7.9%** in 2021 compared to 2019. Meanwhile, Cotai's EUI in October 2021 has decreased by **2.7%** compared to same month in 2019 and decreased by **2.9%** in 2021 compared to 2019.

Table 7. Actual Electricity Consumption Intensity (EUI) Comparison

Monthly Period	Macau			Cotai		
	Year 2019 Actual EUI (kWh/m²)	Year 2021 Actual EUI (kWh/m²)	% Change	Year 2019 Actual EUI (kWh/m²)	Year 2021 Actual EUI (kWh/m²)	% Change
January	24.4	20.4	-16.3%	24.5	21.8	-10.7%
February	23.0	19.8	-14.0%	22.9	20.4	-11.0%
March	25.4	23.3	-8.4%	25.0	24.6	-1.6%
April	27.4	23.8	-13.0%	27.3	25.4	-6.8%
May	30.0	28.9	-3.6%	29.2	30.6	+4.6%
June	32.1	28.7	-10.4%	31.0	29.7	-4.2%
July	33.0	30.8	-6.8%	32.0	31.7	-1.0%
August	32.1	29.8	-7.2%	31.2	30.5	-2.3%
September	29.6	30.3	+2.5%	28.8	30.0	+4.1%
October	28.1	26.8	-4.8%	28.0	27.3	-2.7%
November						
December						
Average	**28.5**	**26.3**	**-7.9%**	**28.0**	**27.2**	**-2.9%**

1.2 Water Use Intensity (WUI)

Macau's water uses intensity (WUI) in October 2021 decreased by **33.2%** compared to the same month in 2019 and decreased by **14.8 %** in 2021 compared to 2019. Meanwhile, Cotai's WUI in October 2021 decreased by **34.1%** compared to the same month in 2019 and decreased by **16.6%** in 2021 compared to 2019.

Table 8. Actual Water Consumption Intensity (WUI) Comparison

Monthly Period	Macau			Cotai		
	Year 2019 Actual WUI (m^3/m^2)	Year 2021 Actual WUI (m^3/m^2)	% Change	Year 2019 Actual WUI (m^3/m^2)	Year 2021 Actual WUI (m^3/m^2)	% Change
January	0.22	0.19	-13.2%	0.16	0.12	-26.2%
February	0.22	0.19	-11.9%	0.17	0.14	-22.0%
March	0.21	0.20	-2.5%	0.17	0.15	-11.8%
April	0.24	0.24	-2.6%	0.20	0.18	-10.9%
May	0.24	0.25	+2.6%	0.20	0.21	+5.7%
June	0.27	0.21	-22.2%	0.21	0.18	-16.6%
July	0.28	0.22	-23.1%	0.23	0.19	-18.6%
August	0.28	0.20	-28.9%	0.23	0.17	-26.3%
September	0.24	0.23	-3.9%	0.19	0.19	-2.6%
October	0.27	0.18	-33.2%	0.22	0.15	-34.1%
November						
December						
Average	**0.25**	**0.21**	**-14.8%**	**0.20**	**0.17**	**-16.6%**

2. MGM Macau Performance

2.1 Electricity Consumption

The actual electricity consumption and cost in October 2021 decreased by **4.8%** and **2.6%** respectively compared to same month in 2019. To date, MGM Macau has decreased the actual electricity consumption and associated costs by **7.9%** and **8.9%** respectively.

Table 9. Macau - Summary of Actual Monthly Electricity Consumption and Cost

Monthly Period	Actual Electricity Consumption (kWh)			Actual Electricity Cost (MOP)			% Change – CDD
	2019	2021	% Change - Actual	2019	2021	% Change - Actual Cost	
January	5,151,480	4,310,160	-16.3%	6,038,725	4,717,260	-21.9%	-18.5%
February	4,870,560	4,190,760	-14.0%	5,728,930	4,591,170	-19.9%	-3.8%
March	5,381,640	4,931,880	-8.4%	6,311,830	5,387,480	-14.6%	+39.0%
April	5,796,240	5,042,160	-13.0%	6,681,548	5,808,340	-13.1%	-5.5%
May	6,349,920	6,122,640	-3.6%	7,312,310	7,034,530	-3.8%	+43.8%
June	6,780,240	6,072,960	-10.4%	8,582,460	7,681,070	-10.5%	-4.4%
July	6,984,480	6,506,760	-6.8%	8,752,550	8,217,570	-6.1%	+1.4%
August	6,799,320	6,306,840	-7.2%	8,526,375	7,960,170	-6.6%	-4.7%
September	6,263,400	6,418,920	+2.5%	7,853,170	8,118,840	+3.4%	+7.9%
October	5,949,120	5,665,560	-4.8%	6,694,080	6,517,390	-2.6%	-16.0%
November							
December							
Total/Average	**60,326,400**	**55,568,640**	**-7.9%**	**72,481,977**	**66,033,820**	**-8.9%**	**+3.2%**

Important note: *Cooling Degree Days (CDDs) is the main variable taken into consideration on Electricity consumption normalization.*

Projects on-going/completed which contributed to electricity consumption decrease below:

- Outdoor billboards lighting equipment upgrades
- Implement PICCV in AHUs (17 units)
- Implement PICCV in Ball Room/Foyer AHUs (3 units)
- Implement Static Pressure Set-Point Reset in AHUs (10 units)
- Implement Motor Efficiency Upgrades (24 units)
    - Total 9 units was completed till September 2021

Table 10 presents the comparison of electricity rates in the year of 2019 and 2021. The electricity rate in October 2021 increased by **2.2%** compared to same month in 2019.

Table 10. Macau - Summary of Actual Monthly Electricity Rate

Monthly Period	Year 2019 – Electricity Rate (MOP/kWh)	Year 2021 – Electricity Rate (MOP/kWh)	% Change – Electricity Rate
January	1.17	1.09	-6.6%
February	1.18	1.10	-6.9%
March	1.17	1.09	-6.9%
April	1.15	1.15	-0.1%
May	1.15	1.15	-0.2%
June	1.27	1.26	-0.1%
July	1.25	1.26	+0.8%
August	1.25	1.26	+0.6%
September	1.25	1.26	+0.9%
October	1.13	1.15	+2.2%
November			
December			
Total/Average	**1.20**	**1.18**	**-1.6%**

Table 11 summarizes the monthly normalized electricity consumption. The normalized electricity consumption in October 2021 decreased by **5.6%** compared to same month in 2019.

Table 11. Macau - Summary of Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2021 Actual Electricity Consumption (kWh)	% Change - Normalized
January	4,763,370	4,310,160	-9.5%
February	5,024,953	4,190,760	-16.6%
March	5,398,600	4,931,880	-8.6%
April	5,765,334	5,042,160	-12.5%
May	6,578,889	6,122,640	-6.9%
June	6,488,142	6,072,960	-6.4%
July	6,770,179	6,506,760	-3.9%
August	6,594,158	6,306,840	-4.4%
September	6,734,744	6,418,920	-4.7%
October	6,004,158	5,665,560	-5.6%
November			
December			
Total/Average	**60,122,527**	**55,568,640**	**-7.6%**

2.2 LPG Consumption

LPG consumption can be broken down into three (3) categories: boiler, hotel upper zone DHW and kitchens/restaurants. Since the boiler and restaurants LPG utilization cover over 95% of the overall site LPG consumption, Energenz focuses the analysis on those two areas.

According to Table 12, the actual LPG consumption and cost in October 2021 increased by **20.1%** and **56.4%** respectively compared to same month in 2019. To date MGM Macau has increased the actual LPG consumption by **1.2%** and associated costs increased by **27.0%**.

Table 12. Macau - Summary of Actual Monthly LPG Consumption and Cost

Monthly Period	Actual LPG Consumption (kg)			Actual LPG Cost (MOP)			% Change – Min OAT
	2019	2021	% Change - Actual	2019	2021	% Change - Actual Cost	
January	94,330	121,398	+28.7%	706,171	947,708	+34.2%	-29.9%
February	72,191	74,797	+3.6%	415,818	632,778	+52.2%	+10.1%
March	78,807	84,131	+6.8%	453,930	711,751	+56.8%	+6.4%
April	81,430	77,702	-4.6%	522,530	652,357	+24.8%	-2.5%
May	84,021	71,294	-15.1%	644,444	578,906	-10.2%	+20.1%
June	78,030	62,104	-20.4%	598,488	504,285	-15.7%	-4.6%
July	73,706	71,398	-3.1%	533,791	579,753	+8.6%	+0.4%
August	79,296	74,204	-6.4%	507,497	698,258	+37.6%	-1.2%
September	75,413	74,426	-1.3%	482,646	700,353	+45.1%	+5.0%
October	76,719	92,146	+20.1%	592,481	926,633	+56.4%	-21.9%
November							
December							
Total/Average	**793,945**	**803,600**	**+1.2%**	**5,457,795**	**6,932,782**	**+27.0%**	**-1.1%**

Important note: *Minimum Outdoor Air Temperature (OAT) is the main variable taken into consideration on LPG consumption normalization.*

Energenz summarized the key items which could result in LPG consumption increase below:

- Boilers LPG consumption increased by 30.8% in October 2021 because steam to water heat exchanger has been switched on for dehumidification at low zone guest room (L5 – L17) from 14 July 2021.
- Hot water supply for FOH and BOH toilets has been enabled on 22 October 2021.
- LPG consumption from Kitchen and Restaurants decreased by 11.5% in October 2021 compared with the same month in 2019.

Table 13 presents the comparison of LPG rates in the year of 2019 and 2021. The LPG rate in October 2021 increased by **30.2%** compared with the same month in 2019.

Table 13. Macau - Summary of Actual Monthly LPG Rate

Monthly Period	Year 2019 – LPG Rate (MOP/kg)	Year 2021 – LPG Rate (MOP/ kg)	% Change – LPG Rate
January	7.49	7.81	+4.3%
February	5.76	8.46	+46.9%
March	5.76	8.46	+46.9%
April	6.42	8.40	+30.8%
May	7.67	8.12	+5.9%
June	7.67	8.12	+5.9%
July	7.24	8.12	+12.1%
August	6.40	9.41	+47.0%
September	6.40	9.41	+47.0%
October	7.72	10.06	+30.2%
November			
December			
Total/Average	**6.85**	**8.64**	**+26.0%**

Table 14 summarizes the monthly normalized LPG consumption. The normalized consumption in October 2021 increased by **26.1%** compared with the actual consumption in 2021.

Table 14. Macau - Summary of Normalized Monthly LPG Consumption

Monthly Period	2019 Normalized LPG Consumption (kg)	2021 Actual LPG Consumption (kg)	% Change – Normalized
January	116,973	121,398	+3.8%
February	83,731	74,797	-10.7%
March	81,839	84,131	+2.8%
April	71,748	77,702	+8.3%
May	74,993	71,294	-4.9%
June	72,822	62,104	-14.7%
July	77,363	71,398	-7.7%
August	74,397	74,204	-0.3%
September	79,156	74,426	-6.0%
October	73,090	92,146	+26.1%
November			
December			
Total/Average	**806,112**	**803,600**	**-0.3%**

2.3 ISO50001 Energy Consumption Record

The Macau ISO50001 energy performance target is to achieve 13% normalized energy savings by 2025 compared to 2019 baseline. Table 15 below summarizes the site monthly energy consumption (Electricity and LPG combined) compared to 2019 baseline. To date, normalized energy saving achieved up to October 2021 is **6.5%**.

Table 15. Macau ISO 50001 Normalized Monthly Energy Consumption

Monthly Period	2019 Normalized Energy Consumption (GJ)	2021 Actual Energy Consumption (GJ)	% Change - Normalized (2021 Vs 2019)
January	22,447	21,016	-6.4%
February	21,883	18,475	-15.6%
March	23,142	21,566	-6.8%
April	24,005	21,672	-9.7%
May	27,081	25,271	-6.7%
June	26,656	24,676	-7.4%
July	27,877	26,659	-4.4%
August	27,109	26,066	-3.8%
September	27,831	26,480	-4.9%
October	24,926	24,570	-1.4%
November			
December			
Total/Average	**252,958**	**236,450**	**-6.5%**

Important note: *Factors used for fuels' unit conversion: 1) Electricity (0.0036) and LPG (0.0453)*

2.4 Water Consumption

Based on the existing water metering system, the overall site water consumption can be broken down into four (4) categories: Kitchen and Restaurants; Hotel/Tower; Fire Services; and Chiller Plant & miscellaneous[2].

According to Table 16, the actual water consumption in October 2021 decreased by **33.2%** compared to same month in 2019. To date, MGM Macau has decreased actual water consumption and associated costs by **14.8%** and **15.0%** respectively.

Table 16. Macau - Summary of Actual Monthly Water Consumption and Cost

Monthly Period	Actual Water Consumption (m^3)			Actual Water Cost (MOP)			Normalization Factors	
	2019	2021	% Change Actual	2019	2021	% Change Actual Cost	% Change – CDD	% Change – Food Covers
January	46,006	39,952	-13.2%	356,970	299,330	-16.1%	-18.5%	-55.0%
February	46,637	41,079	-11.9%	361,510	317,780	-12.1%	-3.8%	-47.7%
March	44,276	43,168	-2.5%	342,840	334,000	-2.6%	+39.0%	-36.4%
April	51,156	49,819	-2.6%	402,280	386,600	-3.9%	-5.5%	-35.8%
May	51,800	53,145	+2.6%	401,890	420,070	+4.5%	+43.8%	-35.9%
June	57,243	44,508	-22.2%	444,090	345,320	-22.2%	-4.4%	-44.0%
July	60,099	46,227	-23.1%	466,230	358,640	-23.1%	+1.4%	-32.0%
August	60,050	42,681	-28.9%	465,790	331,220	-28.9%	-4.7%	-54.6%
September	50,479	48,499	-3.9%	391,440	376,290	-3.9%	+7.9%	-26.8%
October	57,507	38,411	-33.2%	446,140	298,130	-33.2%	-16.0%	-44.2%
November								
December								
Total/Average	**525,253**	**447,489**	**-14.8%**	**4,079,180**	**3,467,380**	**-15.0%**	**+3.2%**	**-41.7%**

Important note: *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization.*

Energenz summarized the key items which could result in water consumption decrease below:

- Hotel Tower water consumption decreased by 46.5% compared to same month in 2019. Moreover, the guest occupancy decreased by 34.6% compared to same month in 2019.
- Podium water consumption for October 2021 decreased by 34.6% compared to the same month in 2019. Moreover, the CDD decreased by 16.0% compared to the same month in 2019.

[2] Miscellaneous includes the water consumption from toilets, laundry, etc.

Table 17 presents the comparison of water rates in the year of 2019 and 2021. The water rate in October 2021 slightly increased by **0.05%** compared to same month in 2019. The average utility rate of water has slightly decreased by **0.3%** compared to 2019.

Table 17. Macau - Summary of Actual Monthly Water Rates

Monthly Period	Year 2019 – Water Rate (MOP/m^3)	Year 2021 – Water Rate (MOP/m^3)	% Change – Water Rate
January	7.76	7.49	-3.4%
February	7.75	7.74	-0.2%
March	7.74	7.74	-0.1%
April	7.86	7.76	-1.3%
May	7.76	7.90	+1.9%
June	7.76	7.76	+0.01%
July	7.76	7.76	+0.01%
August	7.76	7.76	+0.05%
September	7.75	7.76	+0.1%
October	7.76	7.76	+0.05%
November			
December			
Total/Average	**7.77**	**7.74**	**-0.3%**

Table 18 summarized the monthly normalized water consumption. The normalized water consumption in October 2021 decreased by **9.0%** compared to the actual consumption in 2021.

Table 18. Macau - Summary of Normalized Monthly Water Consumption

Monthly Period	2019 Normalized Water Consumption (m^3)	2021 Actual Water Consumption (m^3)	% Change - Normalized
January	31,258	39,952	+27.8%
February	34,783	41,079	+18.1%
March	39,846	43,168	+8.3%
April	42,227	49,819	+18.0%
May	49,504	53,145	+7.4%
June	46,073	44,508	-3.4%
July	51,711	46,227	-10.6%
August	45,118	42,681	-5.4%
September	48,554	48,499	-0.1%
October	42,203	38,411	-9.0%
November			
December			
Total/Average	**431,276**	**447,489**	**+3.8%**

2.5 Macau Year 2021 Projects

Energenz will perform analysis and coordinate with MGM to implement the following projects in 2021 and reduce the site energy and water consumption which supporting Energenz annual normalized energy savings target of 2%.

Table 19. MGM Macau Year 2021 Projects

Item	Initiatives
Electricity	
1	Smoking Lounge Exhaust Fans Operation Optimization
2	Optimize Winter Operation of Cooling Tower Control Strategy - Fan Control
3	Air Change Rate (ACH) Optimization
4	Implement Motor Efficiency Upgrades (24 units)
5	Implement Static Pressure Set-Point Reset in AHU (10 units)
6	Chiller Plant Optimization: Chilled Water Differential Pressure Set-point Optimization
7	Ball Room VAV Recommissioning/Rectification Work
8	Las Vegas Room (LVR) Gaming Area VAV Recommissioning/Rectification Work
9.1	PICCV Implementation – Ball room/Foyer AHU (3 Units)
9.2	PICCV Implementation – RF AHU (2 Units)
9.3	PICCV Implementation (17 Units)
10	Neon Lighting LED Replacement
Water	
11	Optimize Existing Water Softener (Podium)

3. MGM Cotai Performance

3.1 Electricity

The actual electricity consumption and cost in October 2021 decreased by **2.7%** and **0.3%** respectively compared to the same month in 2019. To date, MGM Cotai has decreased actual electricity consumption and associated costs by **2.6%** and **3.8%** respectively.

Table 20. Cotai - Summary of Actual Monthly Electricity Consumption and Cost

Monthly Period	Actual Electricity Consumption (kWh)			Actual Electricity Cost (MOP)			% Change – CDD
	2019	2021	% Change - Actual	2019	2021	% Change - Actual Cost	
January	8,938,560	8,168,400	-8.6%	10,919,060	9,315,910	-14.7%	-18.5%
February	8,560,200	7,637,640	-10.8%	10,478,873	8,737,740	-16.6%	-3.8%
March	9,350,520	9,197,160	-1.6%	11,429,480	10,499,490	-8.1%	+39.0%
April	10,207,920	9,517,560	-6.8%	12,272,805	11,430,660	-6.9%	-5.5%
May	10,940,160	11,446,440	+4.6%	13,125,310	13,735,880	+4.7%	+43.8%
June	11,610,000	11,122,320	-4.2%	13,938,080	13,340,650	-4.3%	-4.4%
July	11,977,200	11,853,120	-1.0%	14,246,260	14,194,900	-0.4%	+1.4%
August	11,685,960	11,412,960	-2.3%	13,897,965	13,675,130	-1.6%	-4.7%
September	10,774,080	11,220,240	+4.1%	12,829,680	13,451,090	+4.8%	+7.9%
October	10,490,280	10,204,200	-2.7%	12,287,710	12,250,640	-0.3%	-16.0%
November							
December							
Total/Average	**104,534,880**	**101,780,040**	**-2.6%**	**125,425,223**	**120,632,090**	**-3.8%**	**+3.2%**

***Important note**: Cooling Degree Days (CDDs) is the main variable taken into consideration on Electricity consumption normalization. The e-Bus electricity consumption has been included with the presented electricity consumption above.*

Energenz summarized the key items which could result in electricity consumption decrease below:

- B1 Carpark and Loading Bay Lighting Operation Schedule Optimization
- Kitchen Exhaust Fan (KEF) Demand Control Ventilation (DCV) and Hood Control for Banquet, Chun, Mansion and Coast kitchen
- Optimize BOH FCU Space Temperature Set-Point
- Optimize Car Park Ventilation Fans Control

Table 21 presents the monthly e-bus electricity consumption in MGM Cotai. The e-bus electricity consumption in October 2021 decreased by **54.5%** compared to same month in 2019. Please note that consumption related to the e-bus generally increases as compared to 2019 as E-bus operation started in January 2019.

Table 21. Cotai - Summary of Monthly e-bus Electricity Consumption

Monthly Period	E-bus Electricity Consumption (kWh)		
	2019	2021	% Change - Actual
January	28,305	44,839	+58.4%
February	24,007	32,534	+35.5%
March	34,315	34,902	+1.7%
April	36,147	37,412	+3.5%
May	37,240	51,584	+38.5%
June	54,293	56,629	+4.3%
July	66,199	64,410	-2.7%
August	74,208	44,185	-40.5%
September	74,828	40,618	-45.7%
October	67,120	30,528	-54.5%
November			
December			
Total/Average	**496,662**	**437,640**	**-11.9%**

Table 22 presents the comparison of electricity rates in the year of 2019 and 2021. The electricity rate in October 2021 increased by **2.5%** compared to same month in 2019.

Table 22. Cotai - Summary of Actual Monthly Electricity Rates

Monthly Period	Year 2019 – Electricity Rate (MOP/kWh)	Year 2021 – Electricity Rate (MOP/kWh)	% Change – Electricity Rate
January	1.22	1.14	-6.6%
February	1.22	1.14	-6.5%
March	1.22	1.14	-6.6%
April	1.20	1.20	-0.1%
May	1.20	1.20	+0.02%
June	1.20	1.20	-0.1%
July	1.19	1.20	+0.7%
August	1.19	1.20	+0.8%
September	1.19	1.20	+0.7%
October	1.17	1.20	+2.5%
November			
December			
Total/Average	**1.20**	**1.18**	**-1.6%**

Table 23 summarized the monthly normalized electricity consumption. The normalized electricity consumption in October 2021 decreased by **2.6%** compared to same month in 2019.

Table 23. Cotai - Summary of Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2021 Actual Electricity Consumption (kWh)	% Change - Normalized
January	8,594,286	8,168,400	-5.0%
February	8,991,397	7,637,640	-15.1%
March	9,558,635	9,197,160	-3.8%
April	10,115,378	9,517,560	-5.9%
May	11,350,444	11,446,440	+0.8%
June	11,212,680	11,122,320	-0.8%
July	11,640,843	11,853,120	+1.8%
August	11,373,624	11,412,960	+0.3%
September	11,587,049	11,220,240	-3.2%
October	10,477,938	10,204,200	-2.6%
November			
December			
Total/Average	**104,902,274**	**101,780,040**	**-3.0%**

3.2 Natural Gas Consumption

The actual natural gas consumption and cost in October 2021 decreased by **40.2%** and **39.8%** respectively compared to same month in 2019. To date, MGM Cotai has decreased actual natural gas consumption and associated costs by **26.3%** and **26.1%** respectively.

Table 24. Cotai - Summary of Actual Monthly Natural Gas Consumption and Cost

Monthly Period	Actual Natural Gas Consumption (m^3)			Actual Natural Gas Cost (MOP)			% Change – Food Covers
	2019	2021	% Change - Actual	2019	2021	% Change - Actual Cost	
January	112,881	60,643	-46.3%	671,460	363,990	-45.8%	-53.9%
February	99,721	54,221	-45.6%	594,320	325,930	-45.2%	-46.8%
March	112,506	68,066	-39.5%	669,261	407,980	-39.0%	-42.2%
April	107,194	86,866	-19.0%	638,120	518,960	-18.7%	-42.5%
May	103,138	90,102	-12.6%	614,347	537,940	-12.4%	-36.0%
June	87,535	80,606	-7.9%	522,890	482,270	-7.8%	-52.7%
July	82,825	102,473	+23.7%	495,276	610,450	+23.3%	-38.7%
August	79,387	51,837	-34.7%	475,084	311,790	-34.4%	-62.5%
September	76,717	51,460	-32.9%	459,259	309,560	-32.6%	-51.1%
October	82,482	49,326	-40.2%	493,270	296,870	-39.8%	-63.5%
November							
December							
Total/Average	**944,386**	**695,600**	**-26.3%**	**5,633,286**	**4,165,740**	**-26.1%**	**-49.5%**

Important note: *Minimum Outdoor Air Temperature (OAT is the main variable taken into consideration on natural gas consumption normalization.*

Energenz summarized the key items which could result in natural gas consumption decrease below:

- Boiler Natural Gas consumption decreased by 62.1% in October 2021 compared to same month in 2019. As the Heat Pump is used for maintaining DHW temperatures and disinfecting the hot water system instead of Natural Gas Boiler from 6 August 2021.
- Kitchen and Restaurants natural gas consumption, which constitutes majority of the overall site consumption, decreased by 40.7% in October 2021 compared to same month in 2019, while food covers had decreased by 63.5%.

Table 25 presents the comparison of natural gas rates in the year of 2019 and 2021. The natural gas rate in October 2021 increased by **0.6%** compared with the same month in 2019.

Table 25. Cotai - Summary of Actual Monthly Natural Gas Rates

Monthly Period	Year 2019 – Natural Gas Rate (MOP/m³)	Year 2021 – Natural Gas Rate (MOP/m³)	% Change – Natural Gas Rate
January	5.95	6.00	+0.9%
February	5.96	6.01	+0.9%
March	5.95	5.99	+0.8%
April	5.95	5.97	+0.4%
May	5.96	5.97	+0.2%
June	5.97	5.98	+0.2%
July	5.98	5.96	-0.4%
August	5.98	6.01	+0.5%
September	5.99	6.02	+0.5%
October	5.98	6.02	+0.6%
November			
December			
Total/Average	**5.97**	**5.99**	**+0.5%**

3.3 ISO50001 Energy Consumption Record

Cotai ISO50001 energy target is to achieve 13% normalized energy consumption reduction by 2025 compared to 2019 baseline. Based on the official agreement with MGM Cotai, e-bus consumption should be included for the energy consumption tracking as part of the ISO50001 requirement. Table 26 below summarizes the site monthly energy consumption (Electricity and natural gas combined) compared to 2019 baseline. The year-to-date normalized energy saving achieved in October 2021 is **6.2%.**

Table 26. Cotai ISO 50001 Normalized Monthly Energy Consumption

Monthly Period	2019 Normalized Energy Consumption (GJ)	2021 Actual Energy Consumption (GJ)	% Change - Normalized (2021 Vs 2019)
January	35,529	31,511	-11.3%
February	36,539	29,377	-19.6%
March	38,505	35,472	-7.9%
April	40,368	37,277	-7.7%
May	44,558	44,334	-0.5%
June	44,002	42,837	-2.6%
July	45,495	46,227	+1.6%
August	44,554	42,885	-3.7%
September	45,225	42,179	-6.7%
October	41,787	38,447	-8.0%
November			
December			
Total/Average	**416,561**	**390,545**	**-6.2%**

***Important note**: Factors used for fuels' unit conversion: 1) Electricity (0.0036) and Natural Gas (0.0347), e-bus consumption is included*

3.4 Water Consumption

According to Table 27, the actual water consumption in October 2021 decreased by **34.1%** compared to same month in 2019. To date, MGM Cotai has decreased actual water consumption and associated costs by **16.5%** and **16.2%** respectively.

Table 27. Cotai - Summary of Actual Monthly Water Consumption and Cost

Monthly Period	Actual Water Consumption (m^3)			Actual Water Cost (MOP)			Normalization Factors	
	2019	2021	% Change Actual	2019	2021	% Change Actual Cost	% Change – CDD	% Change – Food Covers
January	57,331	43,292	-24.5%	445,450	336,700	-24.4%	-18.5%	-53.9%
February	65,011	50,837	-21.8%	505,050	395,180	-21.8%	-3.8%	-46.8%
March	63,136	55,668	-11.8%	488,590	432,400	-11.5%	+39.0%	-42.2%
April	73,577	65,511	-11.0%	571,320	509,000	-10.9%	-5.5%	-42.5%
May	73,317	77,506	+5.7%	569,350	612,200	+7.5%	+43.8%	-36.0%
June	79,312	66,115	-16.6%	615,790	513,500	-16.6%	-4.4%	-52.7%
July	86,836	70,716	-18.6%	674,210	549,160	-18.5%	+1.4%	-38.7%
August	85,257	62,738	-26.4%	661,940	487,380	-26.4%	-4.7%	-62.5%
September	72,406	70,532	-2.6%	562,250	547,710	-2.6%	+7.9%	-51.1%
October	84,105	55,439	-34.1%	653,030	430,760	-34.0%	-16.0%	-63.5%
November								
December								
Total/ Average	**740,288**	**618,354**	**-16.5%**	**5,746,980**	**4,813,990**	**-16.2%**	**+3.2%**	**-49.5%**

***Important note**: Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization.*

Energenz summarized the key items which could result in water consumption decrease below:

- Hotel Tower & Podium water consumption decreased by 41.3% in October 2021 compared to same month in 2019. Moreover, guest occupancy decreased 66.1% in October 2021 compared to same month in 2019.

Table 28 presents the comparison of water rates in the year of 2019 and 2021. The water rate in October 2021 slightly increased by **0.1%** compared to same month in 2019. The average utility rate of water has increased by **0.2%** compared to 2019.

Table 28. Cotai - Summary of Actual Monthly Water Rates

Monthly Period	Year 2019 – Water Rate (MOP/m^3)	Year 2021 – Water Rate (MOP/m^3)	% Change – Water Rate
January	7.77	7.78	+0.1%
February	7.77	7.77	+0.1%
March	7.74	7.77	+0.4%
April	7.76	7.77	+0.1%
May	7.77	7.90	+1.7%
June	7.76	7.77	+0.03%
July	7.76	7.77	+0.02%
August	7.76	7.77	+0.1%
September	7.77	7.77	0.00%
October	7.76	7.77	+0.1%
November			
December			
Total/Average	**7.76**	**7.78**	**+0.2%**

Table 29 summarized the monthly normalized water consumption. The normalized water consumption in October 2021 decreased by **5.3%** compared to the actual consumption in 2021.

Table 29. Cotai - Summary of Normalized Monthly Water Consumption

Monthly Period	2019 Normalized Water Consumption (m^3)	2021 Actual Water Consumption (m^3)	% Change - Normalized
January	45,293	43,292	-4.4%
February	48,765	50,837	+4.2%
March	55,357	55,668	+0.6%
April	58,684	65,511	+11.6%
May	69,926	77,506	+10.8%
June	63,635	66,115	+3.9%
July	71,521	70,716	-1.1%
August	63,252	62,738	-0.8%
September	66,339	70,532	+6.3%
October	58,567	55,439	-5.3%
November			
December			
Total/Average	**601,338**	**618,354**	**+2.8%**

3.5 Cotai Year 2021 Projects

Energenz will perform analysis and coordinate with MGM to implement the following projects in 2021 and reduce the site energy and water consumption which supporting Energenz annual normalized energy savings target of 2%.

Table 30. MGM Cotai Year 2021 Projects

Item	Initiatives
Electricity	
1	Air Change Rate Optimization
2	B1 Carpark and Loading Bay Lighting Operation Schedule Optimization
3	Kitchen Exhaust Fan (KEF) Demand Control Ventilation (DCV) and Hood Control for Remaining Kitchen's (e.g. Banquet, Chun, Mansion, Coast.)
4	Optimize BOH FCU and VAV Temperature Set-Point
5	Optimize Car Park Ventilation Fans Control
6	Optimize Hotel Tower Circulation Pumps Operation
7	Chiller Plant Optimization: 1. Review Chillers Staging 2. Cooling Tower Fan Speed Optimization 3. Condensing Water Pump Speed Optimization
8	Chilled Water Differential Pressure Set-Point Reset Logic Implementation
9	B2 Domestic Hot Water Heat Exchanger Upgrade for fully utilizing the heat pump
10	Implement Static Pressure & Supply Air Temperature Set-Point Reset in AHU (48 Units)
11	PICCV Implementation (7 Units)
12	Solar PV Panel Implementation
Water	
13	Extension of CDP from ST-CD-02 to ST-CD-01 for Cooling Tower Makeup

Appendix A

Regression Analysis

The 2019 normalized consumption modelling is remained used for tracking the agreed normalized energy savings target of 4% accurately for MGM Macau and MGM Cotai. The following tables show a summary of the selected independent variables for each utility which are used for the normalized consumption modelling in 2021.

MGM Macau

Utilities	2021	
Electricity (kWh)	CDD (Base T=17°C)	
LPG (kg)	Min OAT	
Water (m^3)	CDD (Base T=17°C)	Food Cover

MGM Cotai

Utilities	2021	
Electricity (kWh)	CDD (Base T=17°C)	
Natural Gas (m^3)	Min OAT	
Water (m^3)	CDD (Base T=17°C)	Food Cover

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